VOLUME 4B — QUESTION BANK: VISUAL REASONING (150 Questions)

> All items are **text/ASCII-solvable**, so you can practise without images. Items that would normally be shown as a picture are marked **[VISUAL QUESTION — IMAGE REQUIRED]** and include a precise description, so you can still solve them (or draw the figure yourself first: that is excellent practice). Conventions: coordinates are (x, y), with x measured from the left edge and y from the bottom edge. "Vertical mirror" = a mirror standing upright beside the object (left–right flip). "Water image" = a mirror lying below the object (top–bottom flip).

Sections: A. Mirror & water images (1–20) · B. Rotation & symmetry (21–40) · C. Paper folding & cutting (41–60) · D. Cubes, dice, nets (61–85) · E. Counting figures (86–100) · F. Series, matrices & analogies (101–120) · G. Orthographic views, shadows, perspective (121–135) · H. Spatial relations, gears, maps & mixed (136–150)

Format: Question → **Answer** → explanation → *Concept* · *Diff* · *Why NID* · *Mistake*.

A. Mirror & Water Images (1–20)

VR-1. How many of these capital letters look identical in a vertical mirror: A, B, C, M, T, Y, R, W?
(a) 4 (b) 5 (c) 6 (d) 3
Ans: (b). A, M, T, Y, W are symmetric about a vertical axis; B, C, R are not. *Concept:* vertical symmetry · *Diff:* E · *Why NID:* the basic mirror skill · *Mistake:* counting B (which is symmetric about a horizontal axis, not a vertical one).

VR-2. Which word reads exactly the same in a vertical mirror?
(a) TOOT (b) BOOB (c) DEED (d) NOON
Ans: (a). Two conditions are needed: the letters must be vertically symmetric, and the word must read the same backwards. TOOT meets both. B, D, E and N are not vertically symmetric. *Concept:* mirror words · *Diff:* M · *Why NID:* combines two rules · *Mistake:* checking only the palindrome condition.

VR-3. Which word looks unchanged in a water image (top–bottom flip), letter positions unchanged?
(a) CHOICE (b) MOTHER (c) WATER (d) PAPER
Ans: (a). C, H, O, I and E are all symmetric about a horizontal axis. In a water image the order of letters along the line is NOT reversed. *Concept:* water image · *Diff:* M · *Why NID:* the classic item · *Mistake:* reversing the letter order.

VR-4. A clock shows 4:10. What time does its image in a vertical mirror show?
(a) 7:50 (b) 8:50 (c) 7:10 (d) 8:10
Ans: (a). 11:60 − 4:10 = 7:50. *Concept:* mirror clock · *Diff:* E · *Why NID:* spatial transformation · *Mistake:* answering 8:50 by subtracting the hours and minutes separately from 12:60.

VR-5. In a vertical mirror a clock appears to show 2:45. What is the actual time?
(a) 9:15 (b) 10:15 (c) 9:45 (d) 3:15
Ans: (a). The relation is symmetric: 11:60 − 2:45 = 9:15. *Concept:* mirror clock (inverse) · *Diff:* E · *Why NID:* reverse reasoning · *Mistake:* none.

VR-6. Which digit(s) look unchanged in a vertical mirror (simple digital-style digits)?
(a) 0, 1, 8 (b) 2, 5 (c) 3, 8 (d) 6, 9
Ans: (a). In seven-segment form, 0, 8 and the centred "1" are vertically symmetric. 3 is symmetric only about a horizontal axis. *Concept:* digit symmetry · *Diff:* M · *Why NID:* attention to form · *Mistake:* choosing 3.

VR-7. Which digits look unchanged in a water image (seven-segment style)?
(a) 0, 1, 3, 8 (b) 2, 5, 6 (c) 4, 7 (d) 6, 9
Ans: (a). These are symmetric about a horizontal axis. 1 is a vertical stroke, so it stays the same when flipped top to bottom. *Concept:* horizontal symmetry · *Diff:* M · *Why NID:* precision · *Mistake:* choosing 6/9 (that pair swaps under a 180° rotation, not under a water image).

VR-8. The word "NID" is held in front of a vertical mirror. In the mirror, the first letter you read from the left is:
(a) N (b) mirrored D (c) I (d) mirrored N
Ans: (b). The order reverses, so D (now flipped) appears on the left. *Concept:* order reversal · *Diff:* E · *Why NID:* the standard trap · *Mistake:* keeping the original order.

VR-9. An arrow points to the upper right (north-east). In a vertical mirror it points to:
(a) the upper right (b) the upper left (c) the lower right (d) the lower left
Ans: (b). Left–right flips; up stays up. *Concept:* mirror of direction · *Diff:* E · *Why NID:* transformation logic · *Mistake:* answering (d), which is a 180° rotation.

VR-10. The same north-east arrow in a water image points to:
(a) the upper left (b) the lower right (c) the lower left (d) unchanged
Ans: (b). Up flips to down; right stays right. *Concept:* water image · *Diff:* E · *Why NID:* paired with VR-9 · *Mistake:* none.

VR-11. A figure is reflected in a vertical mirror and then reflected again in a horizontal mirror. The net result equals:
(a) the original (b) a 90° rotation (c) a 180° rotation (d) a vertical-mirror image
Ans: (c). Two perpendicular reflections compose to a 180° rotation. *Concept:* composition of transformations · *Diff:* H · *Why NID:* deeper spatial reasoning · *Mistake:* answering (a); two reflections across the *same* axis would give the original.

VR-12. How many letters of the word "EXHIBIT" look the same in a water image?
(a) 5 (b) 6 (c) 7 (d) 4
Ans: (b). E, X, H, I, B, I are symmetric about a horizontal axis; T is not (its bar would move to the bottom). *Concept:* horizontal symmetry · *Diff:* M · *Why NID:* counting with care · *Mistake:* counting T.

VR-13. Ambulances often have the word written as "ƎƆИA⅃UƎMA" on the bonnet. Why?
(a) For style (b) So drivers ahead read it correctly in their rear-view mirrors (c) It's a printing error (d) It's another language
Ans: (b). *Concept:* applied mirror design · *Diff:* E · *Why NID:* links reasoning and real-world design · *Mistake:* none.

VR-14. In a vertical mirror, a person raises their right hand. The mirror image appears to raise:
(a) its right hand (b) its left hand (c) both hands (d) no hand
Ans: (b). From the mirror figure's own point of view, it is raising its left hand. *Concept:* handedness reversal · *Diff:* M · *Why NID:* spatial empathy · *Mistake:* none.

VR-15. A clock shows 12:20. Its vertical-mirror image shows:
(a) 11:40 (b) 12:40 (c) 11:20 (d) 10:40
Ans: (a). For times between 11:00 and 12:59, use 23:60 − 12:20 = 11:40. Check: the minute hand at "4" (20

min) mirrors to "8" (40 min), and the hour hand just past 12 mirrors to just before 12, which gives 11:40 ✓. *Concept:* mirror clock edge case · *Diff:* H · *Why NID:* precision · *Mistake:* computing 11:60 − 12:20 as a negative number.

VR-16. Which letter looks identical in both a vertical mirror and a water image?
(a) A (b) B (c) X (d) E
Ans: (c). X is symmetric about both axes. *Concept:* double symmetry · *Diff:* E · *Why NID:* basic · *Mistake:* none.

VR-17. A car's number plate reads "MH 01". In a vertical mirror, which characters look the same as themselves (ignoring order)?
(a) M, H, 0, 1 (b) M, H only (c) H, 0 only (d) none
Ans: (a). M, H, 0 and 1 are all vertically symmetric. The string order still reverses. *Concept:* character symmetry · *Diff:* M · *Why NID:* observation · *Mistake:* none.

VR-18. Which pair of letters are mirror images of each other (vertical mirror) in standard lowercase?
(a) b and d (b) b and p (c) m and w (d) n and u
Ans: (a). b and p are water images (top–bottom); n and u are 180° rotations; m and w are related by a 180° rotation or water image. *Concept:* letter transformations · *Diff:* M · *Why NID:* fine discrimination · *Mistake:* choosing b/p.

VR-19. Which lowercase pair are water images of each other?
(a) b and d (b) b and p (c) p and q (d) d and q
Ans: (b). b flipped top to bottom becomes p; d flipped becomes q. *Concept:* water image · *Diff:* M · *Why NID:* pairs with VR-18 · *Mistake:* choosing p/q (those are vertical mirrors).

VR-20. A triangle points to the right (▶). Which transformation makes it point left (◀)? Choose ALL that work: (i) vertical mirror (ii) water image (iii) 180° rotation (iv) 90° rotation
(a) i only (b) i and iii (c) ii and iv (d) iii only
Ans: (b). Both a left–right flip and a half turn reverse the direction of a symmetric arrow-triangle. *Concept:* equivalent transformations for symmetric shapes · *Diff:* M · *Why NID:* flexible thinking · *Mistake:* stopping at one answer.

B. Rotation & Symmetry (21–40)

VR-21. An arrow points North. It is rotated 90° clockwise three times, then 45° anticlockwise. Where does it point?
(a) North-West (b) West (c) South-West (d) North
Ans: (c). 270° clockwise puts it at West. Turning 45° anticlockwise from West gives South-West. *Concept:* rotation arithmetic · *Diff:* M · *Why NID:* mental rotation · *Mistake:* turning 45° in the wrong direction.

VR-22. Which of these words reads the same when the page is rotated 180°? (Use simple sans-serif capitals.)
(a) SWIMS (b) MOM (c) DAD (d) WOW
Ans: (a). A 180° rotation reverses the letter order AND turns each letter upside down. S→S, W→M, I→I, M→W, so S-W-I-M-S becomes S-W-I-M-S ✓. MOM becomes WOW ×, WOW becomes MOM ×, and D upside-down is not D ×. *Concept:* 180° ambigrams · *Diff:* H · *Why NID:* careful verification · *Mistake:* only checking whether the word is a palindrome.

VR-23. The number of lines of symmetry of a regular hexagon is:
(a) 3 (b) 4 (c) 6 (d) 12
Ans: (c). Three lines pass through opposite vertices and three through midpoints of opposite sides. *Concept:* symmetry · *Diff:* E · *Why NID:* geometry sense · *Mistake:* answering 3.

VR-24. How many lines of symmetry does a rectangle (not a square) have?
(a) 1 (b) 2 (c) 4 (d) 0
Ans: (b). The diagonals are NOT lines of symmetry for a non-square rectangle. *Concept:* symmetry · *Diff:* E · *Why NID:* the classic trap · *Mistake:* answering 4.

VR-25. A parallelogram (not a rectangle or rhombus) has how many lines of symmetry and what order of rotational symmetry?
(a) 2 lines, order 2 (b) 0 lines, order 2 (c) 0 lines, order 1 (d) 1 line, order 2
Ans: (b). It has no line symmetry, but a half turn maps it onto itself. *Concept:* line vs rotational symmetry · *Diff:* M · *Why NID:* distinguishing the two kinds · *Mistake:* answering (a).

VR-26. Which letter has rotational symmetry of order 2 but NO line of symmetry?
(a) H (b) S (c) A (d) E
Ans: (b). S looks the same after a half turn and has no mirror axis. H has both kinds of symmetry. *Concept:* symmetry types · *Diff:* M · *Why NID:* discrimination · *Mistake:* choosing H.

VR-27. A five-pointed regular star has how many lines of symmetry?
(a) 5 (b) 10 (c) 2 (d) 1
Ans: (a). *Concept:* symmetry · *Diff:* E · *Why NID:* the star motif in logos · *Mistake:* answering 10.

VR-28. The Ashoka Chakra (24 equally spaced spokes) has rotational symmetry of order:
(a) 12 (b) 24 (c) 6 (d) 48
Ans: (b). *Concept:* rotational symmetry · *Diff:* E · *Why NID:* national symbol + geometry · *Mistake:* none.

VR-29. A square tile shows a single arrow pointing up. The tile is rotated 90° anticlockwise. Where does the arrow point?
(a) Right (b) Left (c) Down (d) Up
Ans: (b). Turning anticlockwise, up goes to left. *Concept:* rotation direction · *Diff:* E · *Why NID:* basics · *Mistake:* rotating clockwise by habit.

VR-30. The capital letter "F" is rotated 90° clockwise. What does it look like?
(a) A horizontal bar along the top, with two short strokes hanging down: one from the RIGHT end and one from the middle
(b) A horizontal bar along the top, with two short strokes hanging down: one from the LEFT end and one from the middle
(c) A horizontal bar along the bottom, with two short strokes pointing up from the right end and the middle
(d) It looks like the letter E
Ans: (a). In F the stem is on the left and the arms point right from its top and middle. Turning 90° clockwise sends the top of the stem to the right, so the stem lies horizontal and the arms now point downward. The top arm hangs from the right end; the middle arm hangs from the middle. (b) is the mirror image of the correct answer, and (c) is the result of an anticlockwise turn followed by a flip. *Concept:* mental rotation · *Diff:* H · *Why NID:* precision · *Mistake:* choosing (b). A rotation can never produce a mirror image. (Check: write F on a sticky note and turn it.)

VR-31. A wheel with 8 equally spaced spokes is rotated by 135°. Does it look identical to the start?
(a) Yes (b) No
Ans: (a). Order-8 symmetry means rotations by multiples of 45° look identical, and 135° = 3 × 45°. *Concept:* rotational symmetry · *Diff:* M · *Why NID:* reasoning with angles · *Mistake:* none.

VR-32. Which shape tiles a floor alone with no gaps?
(a) Regular pentagon (b) Regular octagon (c) Regular hexagon (d) Circle
Ans: (c). Its interior angle of 120° divides 360° exactly. *Concept:* tessellation · *Diff:* E · *Why NID:* pattern design · *Mistake:* choosing the octagon (octagons need extra squares to fill the gaps).

VR-33. A clock's minute hand moves from 12 to 9 going clockwise. Through how many degrees has it turned?
(a) 90° (b) 180° (c) 270° (d) 360°
Ans: (c). *Concept:* angle measurement · *Diff:* E · *Why NID:* rotation · *Mistake:* answering 90° (that is the anticlockwise route).

VR-34. A design is made by rotating a single petal around a centre 6 times at equal intervals. The angle between petals is:
(a) 30° (b) 45° (c) 60° (d) 72°
Ans: (c). 360° ÷ 6 = 60°. *Concept:* radial design · *Diff:* E · *Why NID:* rangoli / mandala construction · *Mistake:* none.

VR-35. Which letter has exactly one horizontal line of symmetry and no vertical one?
(a) A (b) K (c) H (d) T
Ans: (b). A and T have a vertical axis; H has both. *Concept:* symmetry · *Diff:* M · *Why NID:* discrimination · *Mistake:* none.

VR-36. A semicircle has how many lines of symmetry?
(a) 0 (b) 1 (c) 2 (d) infinite
Ans: (b). *Concept:* symmetry · *Diff:* E · *Why NID:* basics · *Mistake:* answering "infinite" by confusing it with a full circle.

VR-37. Which of these figures has rotational symmetry of order 3?
(a) Square (b) Equilateral triangle (c) Rectangle (d) Isosceles triangle
Ans: (b). *Concept:* rotational symmetry · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-38. A 3×3 grid has only its top-left cell shaded. The grid is rotated 90° clockwise, then mirrored left–right. Which cell is shaded now?
(a) top-left (b) top-right (c) bottom-left (d) bottom-right
Ans: (a). A 90° clockwise rotation moves top-left to top-right. The left–right mirror moves top-right back to top-left. *Concept:* composed transformations · *Diff:* M · *Why NID:* multi-step tracking · *Mistake:* doing the steps in the wrong order.

VR-39. In the same 3×3 grid, the shaded cell is top-middle. After a 180° rotation it is:
(a) top-middle (b) bottom-middle (c) middle-left (d) centre
Ans: (b). *Concept:* rotation · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-40. A paper strip has a half-twist and its ends are joined (a Möbius strip). How many sides and edges does it have?
(a) 2 sides, 2 edges (b) 1 side, 1 edge (c) 1 side, 2 edges (d) 2 sides, 1 edge
Ans: (b). *Concept:* topology · *Diff:* M · *Why NID:* curiosity / paper models (the recycling symbol is based on it) · *Mistake:* answering (a).

C. Paper Folding & Cutting (41–60)

VR-41. A square sheet, 8 cm × 8 cm, is folded in half with the left half going onto the right half (fold line x = 4). A hole is punched at (6, 6). Unfolded, where are the holes?
(a) (6,6) only (b) (2,6) and (6,6) (c) (6,2) and (6,6) (d) (2,2) and (6,6)
Ans: (b). Reflecting across x = 4 gives 4 − (6 − 4) = 2, with y unchanged. *Concept:* reflection across a fold · *Diff:* E · *Why NID:* the basic fold · *Mistake:* reflecting y as well.

VR-42. The same 8×8 sheet is folded left onto right (x = 4), then the top onto the bottom (y = 4). One hole is punched at (5, 3). How many holes appear, and where?
(a) 2 (b) 4: (5,3), (3,3), (5,5), (3,5) (c) 4: (5,3), (3,3), (5,1), (3,1) (d) 8
Ans: (b). Unfold in reverse order. First undo the top-onto-bottom fold: (5,3) reflects across y = 4 to (5,5). Then undo the left-onto-right fold: (5,3)→(3,3) and (5,5)→(3,5). *Concept:* reverse unfolding · *Diff:* M · *Why NID:* the core skill · *Mistake:* reflecting across the wrong axis.

VR-43. A sheet is folded in half 4 times, and a hole is punched through all layers once (not on a fold). How many holes appear when unfolded?
(a) 4 (b) 8 (c) 16 (d) 32
Ans: (c). 2^4 = 16. *Concept:* layer doubling · *Diff:* E · *Why NID:* the quick rule · *Mistake:* answering 8.

VR-44. A square sheet is folded along its diagonal from bottom-left to top-right (the top-left triangle goes onto the bottom-right). A hole is punched at (6, 2) on an 8×8 sheet. Where is the second hole after unfolding?
(a) (2,6) (b) (6,6) (c) (2,2) (d) (6,2) only
Ans: (a). Reflection across the line y = x swaps the coordinates: (6,2) → (2,6). *Concept:* diagonal fold · *Diff:* M · *Why NID:* the harder fold type · *Mistake:* treating it as a vertical fold.

VR-45. A rectangle is folded in half once, and a semicircle is cut out with its flat side on the fold line. Unfolded, the hole is:
(a) a semicircle (b) a full circle (c) two semicircles at opposite edges (d) an ellipse
Ans: (b). The cut mirrors across the fold to complete a circle. *Concept:* cut on a fold · *Diff:* E · *Why NID:* paper cutting · *Mistake:* answering (c).

VR-46. A paper is folded in half twice (making 4 layers, folds meeting at one corner). A quarter-circle is cut from the corner where both folds meet. Unfolded, the result is:
(a) a quarter-circle notch (b) a full circular hole in the centre (c) four circles (d) a semicircle on an edge
Ans: (b). The corner where the folds meet is the centre of the original sheet, so four quarter-circles join into a full circle. *Concept:* the snowflake principle · *Diff:* M · *Why NID:* the classic item · *Mistake:* answering (c).

VR-47. A long strip of paper is folded in half 3 times, each fold parallel to the previous one (making 8 layers). It is then cut with ONE straight cut across all layers, parallel to the folds but not on a fold. How many pieces result?
(a) 2 (b) 5 (c) 8 (d) 9
Ans: (d). Each cut through the folded packet separates the strip at every layer, giving 8 cut lines on the unfolded strip and therefore 8 + 1 = 9 pieces. Test with small cases: 1 fold (2 layers) gives 3 pieces, and 2 folds (4 layers) give 5. The pattern is $2^n + 1$. *Concept:* fold-and-cut reasoning · *Diff:* H · *Why NID:* deeper reasoning · *Mistake:* answering 8 (the layers, not the pieces).

VR-48. A paper is folded like a fan (accordion) into 4 equal vertical strips. A small triangle is cut out of the CENTRE of the folded stack (not touching any edge or fold). How many triangular holes appear when unfolded?
(a) 1 (b) 2 (c) 4 (d) 3
Ans: (c). The accordion makes 4 layers, so the cut passes through 4 layers and gives 4 holes. Neighbouring holes are mirror images of each other across the folds. *Concept:* accordion fold · *Diff:* M · *Why NID:* garland and paper-chain designs · *Mistake:* answering 1.

VR-49. Paper doll chains (people holding hands) are made by accordion-folding and cutting a figure whose hands touch the folded edges. Why must the hands touch the folds?
(a) For decoration (b) Uncut fold segments keep the figures connected (c) To save paper (d) There is no

reason
Ans: (b). *Concept:* the fold as a hinge or connection · *Diff:* E · *Why NID:* craft logic · *Mistake:* none.

VR-50. An 8×8 sheet is folded so that the top edge meets the bottom edge (fold at y = 4), then the right edge meets the left edge (fold at x = 4). A hole is punched at (1, 1) in the folded packet, which spans x 0–4 and y 0–4. Unfolded holes:
(a) (1,1), (7,1), (1,7), (7,7) (b) (1,1), (3,1), (1,3), (3,3) (c) (1,1) only (d) (1,1), (1,7)
Ans: (a). Undo the right-onto-left fold: (1,1)→(7,1). Undo the top-onto-bottom fold: each point reflects across y = 4, so (1,1)→(1,7) and (7,1)→(7,7). *Concept:* reverse unfolding · *Diff:* M · *Why NID:* practice · *Mistake:* reflecting across x = 2.

VR-51. Which fold produces a mirror-symmetric shape with a vertical axis when cutting a heart?
(a) Fold vertically and cut half a heart along the fold (b) Fold horizontally (c) Fold diagonally (d) No fold
Ans: (a). *Concept:* symmetric cutting · *Diff:* E · *Why NID:* craft · *Mistake:* none.

VR-52. A sheet is folded in half. A square punch is placed so that it straddles the folded edge (half of the punch over the paper, half hanging off the fold). Unfolded, you see:
(a) two separate square holes (b) one rectangular hole twice as wide as the part of the punch that bit into the paper, centred on the fold line (c) a notch on the outer edge (d) no hole
Ans: (b). The punch cuts a notch into the folded edge. Unfolding mirrors the notch across the fold, joining the two halves into one hole that straddles the fold line. *Concept:* punch on a fold · *Diff:* M · *Why NID:* the typical trap · *Mistake:* answering (a).

VR-53. How many layers result from folding a sheet in half, then in thirds (accordion, 3 panels)?
(a) 5 (b) 6 (c) 8 (d) 9
Ans: (b). 2 × 3 = 6. *Concept:* layer multiplication · *Diff:* E · *Why NID:* reasoning · *Mistake:* adding instead of multiplying.

VR-54. A square paper is folded in half twice to form a smaller square (4 layers). One small circle is punched at the centre of this small square (not on any fold). How many holes appear after unfolding, and how are they arranged?
(a) 4, one at the centre of each quadrant (b) 4, clustered at the centre (c) 1 at the centre (d) 2
Ans: (a). The centre of the folded square maps to the centre of each quarter of the original sheet. *Concept:* mapping folded positions · *Diff:* M · *Why NID:* practice · *Mistake:* answering (b).

VR-55. Origami is the Japanese art of paper folding. Kirigami differs because it:
(a) uses glue (b) involves cutting as well as folding (c) uses cloth (d) uses no folds
Ans: (b). *Concept:* paper craft vocabulary · *Diff:* E · *Why NID:* the studio test · *Mistake:* none.

VR-56. A strip of paper is folded in half 5 times along its length, then unfolded. How many crease lines are there?
(a) 5 (b) 16 (c) 31 (d) 32
Ans: (c). Folding 5 times creates 2^5 = 32 segments, separated by 31 creases. *Concept:* creases = segments − 1 · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 32.

VR-57. An 8×8 square is folded diagonally from the top-left corner to the bottom-right corner (the bottom-left triangle folds onto the top-right triangle). A hole is punched at (6, 6). The second hole after unfolding is at:
(a) (2,2) (b) (6,6) only (c) (2,6) (d) (6,2)
Ans: (a). The fold line runs from (0,8) to (8,0), i.e. x + y = 8. Reflecting (6,6) across it gives (8 − 6, 8 − 6) = (2,2). *Concept:* reflection across the anti-diagonal · *Diff:* H · *Why NID:* advanced folding · *Mistake:* swapping coordinates as if the fold were y = x.

VR-58. A circular paper (a coffee filter) is folded in half, then in half again, then once more, making 8 layers in a narrow wedge. One small hole is punched in the interior of the wedge, not touching any edge or fold. How many holes appear after unfolding?
(a) 4 (b) 8 (c) 16 (d) 2
Ans: (b). 8 layers give 8 holes, arranged in a ring with radial symmetry. *Concept:* radial cutting (snowflakes) · *Diff:* M · *Why NID:* radial design · *Mistake:* answering 4.

VR-59. To make a 6-pointed paper snowflake, the circle or square should be folded so that the wedge angle is:
(a) 90° (b) 60° (c) 30° (d) 45°
Ans: (c). Six-fold symmetry with mirror lines needs 12 identical mirrored wedges, and 360° ÷ 12 = 30°. (A 60° wedge gives 6 identical parts, but the mirror symmetry comes from folding the wedge in half.) *Concept:* radial symmetry construction · *Diff:* H · *Why NID:* deeper craft geometry · *Mistake:* answering 60°.

VR-60. A sheet folded in half twice, at right angles, has a hole punched through all layers. One of the unfolded holes is at the top-left quadrant's centre. Where are the others?
(a) The centres of the other three quadrants (b) Along the top edge only (c) The centre of the sheet only (d) Random
Ans: (a). *Concept:* symmetry about both fold lines · *Diff:* E · *Why NID:* reinforcement · *Mistake:* none.

D. Cubes, Dice & Nets (61–85)

VR-61. A cube has how many edges?
(a) 6 (b) 8 (c) 12 (d) 24
Ans: (c). *Concept:* polyhedra · *Diff:* E · *Why NID:* basics · *Mistake:* answering 8, which is the number of vertices.

VR-62. On a standard die, which number is opposite 3?
(a) 4 (b) 5 (c) 6 (d) 1
Ans: (a). Opposite faces sum to 7. *Concept:* the dice rule · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-63. In the net below, which face is opposite face C?

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   [A]
[B][C][D][E]
   [F]
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(a) A (b) E (c) F (d) B
Ans: (b). In a row of 4, faces with one face between them are opposite: B–D and C–E. A and F (above and below the same face C) are opposite each other. *Concept:* the net rule · *Diff:* M · *Why NID:* standard cube item · *Mistake:* answering D.

VR-64. In the same net, which face is opposite A?
(a) F (b) C (c) E (d) D
Ans: (a). Both attach to face C on opposite sides of it. *Concept:* net · *Diff:* M · *Why NID:* the same net, a new face · *Mistake:* answering C.

VR-65. Can this arrangement fold into a cube?

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[1][2][3]
[4][5][6]
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(a) Yes (b) No
Ans: (b). A 2×3 block has faces that would overlap; no face could close the cube. *Concept:* invalid nets · *Diff:* M · *Why NID:* net validity · *Mistake:* thinking 6 squares are always enough.

VR-66. Can this fold into a cube?

```
[1][2][3][4]
[5]      [6]
```

(1 is above 5; 4 is above 6)
(a) Yes (b) No
Ans: (b). With 4 in a row and the other two squares both hanging on the SAME side, at the two ends, faces 5 and 6 would both fold to the same side of the cube and collide; nothing would close the face opposite their side. Check: in the row 1-2-3-4, opposite pairs are 1–3 and 2–4. Face 5 (under 1) and face 6 (under 4) fold in from the same direction and land on the same face position. *Concept:* invalid net · *Diff:* H · *Why NID:* deeper visualisation · *Mistake:* assuming any 4 + 2 arrangement works.

VR-67. Can this fold into a cube? A row of four squares 1-2-3-4; square 5 is attached ABOVE square 2; square 6 is attached BELOW square 4.

```
   [5]
[1][2][3][4]
         [6]
```

(a) Yes (b) No
Ans: (a). Four in a row with one square above and one below (at any positions along the row) is always a valid net, and there are 6 such variants. *Concept:* the 1-4-1 net family · *Diff:* M · *Why NID:* net validity · *Mistake:* none.

VR-68. For the valid net in VR-67, which face is opposite face 5?
(a) 6 (b) 1 (c) 3 (d) 4
Ans: (a). In 1-4-1 nets, the square above the row and the square below the row are always opposite each other. *Concept:* net rule · *Diff:* M · *Why NID:* follow-up · *Mistake:* none.

VR-69. A 3×3×3 cube painted on all faces is cut into 27 small cubes. How many small cubes have exactly 2 painted faces?
(a) 8 (b) 12 (c) 6 (d) 1
Ans: (b). $12(n-2) = 12 \times 1 = 12$, one on each edge. *Concept:* painted cube · *Diff:* E · *Why NID:* standard · *Mistake:* answering 8, which is the number of corners.

VR-70. A 5×5×5 painted cube is cut into 125 unit cubes. How many have exactly 1 painted face?
(a) 54 (b) 36 (c) 27 (d) 72
Ans: (a). $6(n-2)^2 = 6 \times 9 = 54$. *Concept:* painted cube · *Diff:* M · *Why NID:* formula use · *Mistake:* answering 27, which is the unpainted count.

VR-71. A 6×6×6 painted cube is cut into unit cubes. How many have NO paint?
(a) 64 (b) 125 (c) 216 (d) 96
Ans: (a). $(6-2)^3 = 64$. *Concept:* painted cube · *Diff:* M · *Why NID:* formula · *Mistake:* none.

VR-72. A cuboid 3×4×5 is painted on all faces and cut into unit cubes. How many unit cubes have no paint?
(a) 6 (b) 12 (c) 24 (d) 60
Ans: (a). The interior is $(3-2)(4-2)(5-2) = 1 \times 2 \times 3 = 6$. *Concept:* painted cuboid · *Diff:* M · *Why NID:* the extension · *Mistake:* using a cube formula.

VR-73. In the same 3×4×5 cuboid, how many unit cubes have exactly 2 painted faces?
(a) 24 (b) 12 (c) 20 (d) 36
Ans: (a). Edge cubes excluding corners: 4[(3−2) + (4−2) + (5−2)] = 4 × 6 = 24. *Concept:* painted cuboid · *Diff:* H · *Why NID:* advanced · *Mistake:* none.

VR-74. What is the minimum number of straight cuts needed to divide a cube into 27 equal smaller cubes (pieces may not be rearranged)?
(a) 6 (b) 9 (c) 27 (d) 3
Ans: (a). Two cuts in each of the three directions. *Concept:* spatial division · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

VR-75. Two views of the same die: View 1 shows 1 on top, 2 in front. View 2 shows 1 on top, 3 in front. Which statement must be true?
(a) 2 is opposite 3 (b) 1 is opposite 6 (on a standard die) (c) 2 is opposite 4 (d) 3 is opposite 1
Ans: (b). On a standard die, opposite faces sum to 7. The two views alone show only that 2 and 3 are both adjacent to 1, so (a) cannot be concluded. *Concept:* dice logic · *Diff:* M · *Why NID:* inference discipline · *Mistake:* concluding (a).

VR-76. A die is rolled one quarter turn forward (away from you) on a table, starting with 1 on top and 2 facing you. After the roll, what is on top? (Standard die.)
(a) 2 (b) 5 (c) 6 (d) 3
Ans: (a). Rolling forward brings the front face up to the top. *Concept:* rolling dice · *Diff:* M · *Why NID:* mental simulation · *Mistake:* answering 5.

VR-77. A cube has faces coloured Red, Blue, Green, Yellow, Black, White. Red is opposite Blue, Green is opposite Yellow. Which colour is opposite White?
(a) Red (b) Black (c) Green (d) Yellow
Ans: (b). The remaining pair must be opposite. *Concept:* elimination · *Diff:* E · *Why NID:* logic · *Mistake:* none.

VR-78. How many different nets (unfolded shapes) does a cube have?
(a) 6 (b) 8 (c) 11 (d) 12
Ans: (c). *Concept:* nets · *Diff:* M · *Why NID:* the classic fact · *Mistake:* none.

VR-79. Which solid has a net made ONLY of squares?
(a) Cube (b) Square pyramid (c) Triangular prism (d) Cylinder
Ans: (a). A square pyramid needs 4 triangles plus a square, a triangular prism needs 2 triangles plus 3 rectangles, and a cylinder needs 2 circles plus a rectangle. *Concept:* nets of solids · *Diff:* E · *Why NID:* packaging dielines · *Mistake:* choosing the square pyramid because of its square base.

VR-80. The net of a cylinder consists of:
(a) 2 circles and 1 rectangle (b) 3 circles (c) 2 rectangles and 1 circle (d) 1 circle and 1 triangle
Ans: (a). *Concept:* nets · *Diff:* E · *Why NID:* packaging · *Mistake:* none.

VR-81. The net of a triangular prism contains:
(a) 2 triangles and 3 rectangles (b) 3 triangles and 2 rectangles (c) 5 triangles (d) 6 squares
Ans: (a). *Concept:* nets · *Diff:* E · *Why NID:* the Toblerone-style packaging · *Mistake:* none.

VR-82. A tetrahedron has how many faces, edges and vertices?
(a) 4, 6, 4 (b) 4, 4, 4 (c) 6, 12, 8 (d) 5, 8, 5
Ans: (a). Check with Euler's formula V − E + F = 4 − 6 + 4 = 2 ✓. *Concept:* polyhedra · *Diff:* M · *Why NID:* form vocabulary · *Mistake:* none.

VR-83. Using Euler's formula (V − E + F = 2), a polyhedron with 12 vertices and 30 edges has how many faces?
(a) 18 (b) 20 (c) 22 (d) 16
Ans: (b). 12 − 30 + F = 2, so F = 20. (This is an icosahedron.) *Concept:* Euler · *Diff:* M · *Why NID:* logical geometry · *Mistake:* none.

VR-84. A stack of cubes has 3 layers: the bottom is 3×3 = 9 cubes, the middle 2×2 = 4, and the top is 1. How many cubes are in total, and how many are completely hidden from view (from all sides and the top, assuming the bottom isn't visible)?
(a) 14 total; 1 hidden (b) 14 total; 0 hidden (c) 13 total; 1 hidden (d) 14 total; 2 hidden
Ans: (a). 9 + 4 + 1 = 14. The bottom-centre cube is covered by the middle layer above and surrounded on all four sides by bottom-layer cubes, so it is hidden. In the middle layer every cube is on the edge, so all are visible. *Concept:* stacked solids · *Diff:* M · *Why NID:* spatial counting · *Mistake:* answering 0 hidden.

VR-85. A 2×2×2 cube made of 8 unit cubes is painted red on the outside. Then one unit cube is removed. How many unit faces of the remaining structure show red on the outside?
(a) 21 (b) 24 (c) 18 (d) 20
Ans: (a). Originally there are 24 red unit faces (6 faces × 4). The removed cube was a corner cube carrying 3 red faces, so 24 − 3 = 21 red faces remain. The removal exposes 3 new, unpainted inner faces of its neighbours. *Concept:* surface reasoning · *Diff:* H · *Why NID:* deeper spatial thinking · *Mistake:* forgetting that the exposed faces are unpainted.

E. Counting Figures (86–100)

VR-86. How many squares (of all sizes) are in a 4×4 grid of unit squares?
(a) 16 (b) 25 (c) 30 (d) 20
Ans: (c). 16 + 9 + 4 + 1 = 30. *Concept:* counting squares · *Diff:* M · *Why NID:* observation · *Mistake:* answering 16.

VR-87. How many rectangles (including squares) are in a 3×3 grid?
(a) 14 (b) 36 (c) 9 (d) 18
Ans: (b). C(4,2) × C(4,2) = 6 × 6 = 36. *Concept:* rectangle counting · *Diff:* M · *Why NID:* combinatorial reasoning · *Mistake:* answering 14, which counts squares only.

VR-88. A square with both diagonals drawn contains how many triangles?
(a) 4 (b) 6 (c) 8 (d) 10
Ans: (c). 4 small triangles plus 4 half-squares, each made of 2 small triangles. *Concept:* triangle counting · *Diff:* E · *Why NID:* standard · *Mistake:* answering 4.

VR-89. A square with both diagonals AND both midlines (the lines joining midpoints of opposite sides) drawn: how many triangles?
(a) 8 (b) 12 (c) 16 (d) 20
Ans: (c). There are 8 smallest triangles. Two adjacent smallest triangles meeting across a midline make 4 medium triangles (each with the centre and two adjacent corners). The diagonals split the square into halves, giving 4 large triangles. 8 + 4 + 4 = 16. *Concept:* systematic counting · *Diff:* H · *Why NID:* the classic tough item · *Mistake:* missing the medium triangles.

VR-90. An equilateral triangle is divided into 9 small equilateral triangles (3 per side). How many triangles in total?
(a) 9 (b) 10 (c) 13 (d) 16
Ans: (c). Upward triangles: 6 of size 1, 3 of size 2 and 1 of size 3 make 10. There are 3 downward triangles of size 1. Total 13. *Concept:* triangular grids · *Diff:* H · *Why NID:* systematic counting · *Mistake:* forgetting the downward triangles.

VR-91. A triangle has 5 lines from its apex to its base (including the two sides). How many triangles?
(a) 5 (b) 10 (c) 15 (d) 20
Ans: (b). Any two of the 5 lines, together with the base, form a triangle: C(5,2) = 10. *Concept:* combinations · *Diff:* M · *Why NID:* the formula approach · *Mistake:* none.

VR-92. The same figure as VR-91 with one horizontal line crossing all 5 lines (parallel to the base). How many triangles?
(a) 10 (b) 15 (c) 20 (d) 25
Ans: (c). Each horizontal "base" (the actual base plus the crossing line) gives 10 triangles, so 2 × 10 = 20. *Concept:* extending the formula · *Diff:* H · *Why NID:* deeper · *Mistake:* answering 15.

VR-93. How many squares are in a 2×3 grid?
(a) 6 (b) 8 (c) 9 (d) 11
Ans: (b). 6 unit squares plus 2 squares of size 2×2. *Concept:* counting in non-square grids · *Diff:* M · *Why NID:* care · *Mistake:* answering 6.

VR-94. How many straight lines are needed to draw a 5-pointed star (pentagram) without lifting the pencil?
(a) 4 (b) 5 (c) 6 (d) 10
Ans: (b). *Concept:* figure construction · *Diff:* E · *Why NID:* observation · *Mistake:* answering 10, which counts the outer edges.

VR-95. How many triangles are in a pentagram (5-pointed star drawn with 5 straight lines)?
(a) 5 (b) 10 (c) 15 (d) 35
Ans: (b). Every set of 3 of the 5 lines meets in pairs at points that lie on the drawn segments, so each set of 3 lines bounds a triangle: C(5,3) = 10. These are 5 small "tip" triangles and 5 larger triangles (each made of one tip plus part of the inner pentagon). *Concept:* systematic counting · *Diff:* H · *Why NID:* the classic puzzle · *Mistake:* answering 5 (only the tips).

VR-96. [VISUAL QUESTION — IMAGE REQUIRED] *Figure description:* a large rectangle divided by one vertical line into two equal halves; in the left half, one diagonal from top-left to bottom-right. How many triangles?
(a) 1 (b) 2 (c) 3 (d) 4
Ans: (b). The diagonal splits the left half into 2 triangles. The right half has no diagonal. *Concept:* careful counting · *Diff:* E · *Why NID:* attention · *Mistake:* imagining extra lines.

VR-97. How many cubes are in a staircase made of columns of heights 1, 2, 3, 4 (in a single row)?
(a) 8 (b) 10 (c) 12 (d) 16
Ans: (b). 1 + 2 + 3 + 4 = 10. *Concept:* counting solids · *Diff:* E · *Why NID:* spatial counting · *Mistake:* none.

VR-98. How many diagonals does an octagon have?
(a) 16 (b) 20 (c) 24 (d) 28
Ans: (b). n(n − 3)/2 = 8 × 5 / 2 = 20. *Concept:* diagonals · *Diff:* M · *Why NID:* geometry · *Mistake:* none.

VR-99. How many circles of diameter 2 cm fit in a single row inside a rectangle 10 cm long and 2 cm wide (without overlapping)?
(a) 4 (b) 5 (c) 10 (d) 20
Ans: (b). *Concept:* packing · *Diff:* E · *Why NID:* layout logic · *Mistake:* none.

VR-100. How many squares are in a 5×5 grid?
(a) 25 (b) 50 (c) 55 (d) 60
Ans: (c). 25 + 16 + 9 + 4 + 1 = 55. *Concept:* counting · *Diff:* M · *Why NID:* the formula · *Mistake:* none.

F. Series, Matrices & Visual Analogies (101–120)

VR-101. Series of arrows: ↑ → ↓ ← ↑ → ? What comes next?
(a) ↑ (b) ↓ (c) ← (d) →
Ans: (b). The arrow rotates 90° clockwise each step. *Concept:* rotation series · *Diff:* E · *Why NID:* pattern recognition · *Mistake:* none.

VR-102. Series: ● ○○ ●●● ○○○○ ? What comes next?
(a) ●●●●● (b) ○○○○○ (c) ●●●● (d) ○●○●○
Ans: (a). The count increases by 1 each step and the fill alternates between filled and empty. *Concept:* two-rule series · *Diff:* E · *Why NID:* tracking multiple rules · *Mistake:* none.

VR-103. A shaded quadrant in a circle moves: top-left, top-right, bottom-right, bottom-left, ? Next:
(a) top-left (b) bottom-right (c) centre (d) top-right
Ans: (a). It moves clockwise and cycles. *Concept:* cyclic series · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-104. Sequence of polygons: triangle, square, pentagon, ?, heptagon. Missing:
(a) circle (b) hexagon (c) octagon (d) rhombus
Ans: (b). The number of sides increases by 1. *Concept:* series · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-105. Matrix (number of dots in each cell):

```
1  2  3
2  4  6
3  6  ?
```

(a) 8 (b) 9 (c) 12 (d) 7
Ans: (b). Each cell is row × column, so 3 × 3 = 9. *Concept:* matrix rule · *Diff:* E · *Why NID:* matrix reasoning · *Mistake:* answering 8.

VR-106. Matrix of shapes; each row contains one circle, one square and one triangle, with fill varying:

```
○  ■  △
■  △  ○
△  ○  ?
```

(Row 1: empty circle, filled square, empty triangle. Row 2: filled square, empty triangle, empty circle.)
Missing shape:
(a) a square (b) a circle (c) a triangle (d) a star
Ans: (a). Each shape appears once in each row and column (a Latin square), so row 3 needs the square. *Concept:* Latin-square logic · *Diff:* M · *Why NID:* matrix · *Mistake:* none.

VR-107. Analogy: a square is to a cube as a circle is to a:
(a) cylinder (b) sphere (c) cone (d) ring
Ans: (b). The 2D shape maps to its 3D counterpart, which has the same outline from every direction. *Concept:* 2D→3D analogy · *Diff:* E · *Why NID:* form vocabulary · *Mistake:* answering cylinder (which comes from extruding a circle).

VR-108. Analogy: a small circle inside a big square is to a big circle inside a small square as a small triangle inside a big circle is to:
(a) a small circle inside a big triangle (b) a big circle with a small triangle inside it (c) a small triangle inside a big square (d) two triangles

Ans: (a). The rule swaps which shape is outside and which is inside. Inner small triangle with outer big circle becomes inner small circle with outer big triangle. *Concept:* visual analogy (swap) · *Diff:* M · *Why NID:* analogical thinking · *Mistake:* choosing (b), which is just the original figure.

VR-109. Odd one out: cube, cylinder, sphere, square.
(a) cube (b) cylinder (c) sphere (d) square
Ans: (d). It is the only 2D shape. *Concept:* classification · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-110. Odd one out: ⬠ (pentagon), ⬡ (hexagon), △ (triangle), ◯ (circle).
(a) pentagon (b) hexagon (c) triangle (d) circle
Ans: (d). It is the only shape with no straight edges or vertices. *Concept:* classification · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-111. Series: a clock face shows 12:00, 12:15, 12:45, 1:30, ? What comes next?
(a) 2:15 (b) 2:30 (c) 2:00 (d) 3:00
Ans: (b). The intervals are +15, +30, +45, so the next is +60 min: 1:30 + 1:00 = 2:30. *Concept:* increasing intervals · *Diff:* M · *Why NID:* visual series · *Mistake:* answering 2:15.

VR-112. Series of figures with sides: 3, 4, 6, 9, ? (triangle, square, hexagon, nonagon…). Next number of sides:
(a) 12 (b) 13 (c) 14 (d) 11
Ans: (b). The differences are 1, 2, 3, so the next is +4: 9 + 4 = 13. *Concept:* series · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 12.

VR-113. Matrix: each row's third figure is formed by overlaying the first two and removing the lines they share (an XOR rule). Row: Figure 1 = a square with a vertical midline; Figure 2 = a square with a horizontal midline. Figure 3 = ?
(a) an empty square (b) a square with both midlines (a plus sign inside) (c) only a plus sign (d) a square with diagonals
Ans: (c). The outer square is common to both figures and is removed. The midlines are not shared, so they remain. *Concept:* superimposition rules · *Diff:* H · *Why NID:* matrix reasoning · *Mistake:* answering (b), which is the plain-overlay (OR) rule.

VR-114. Using the plain overlay rule (OR) instead, Figure 3 in VR-113 would be:
(a) an empty square (b) a square with both midlines (c) a plus sign only (d) diagonals
Ans: (b). *Concept:* rule variants · *Diff:* M · *Why NID:* rule precision · *Mistake:* none.

VR-115. Series: a dot moves around the 8 outer cells of a 3×3 grid, skipping one cell each time (moving 2 positions clockwise). Starting at the top-left corner, where is it after 3 moves?
(a) bottom-left (b) bottom-right (c) top-right (d) middle-left
Ans: (a). Number the outer ring clockwise from top-left: 1 TL, 2 TM, 3 TR, 4 MR, 5 BR, 6 BM, 7 BL, 8 ML. Starting at 1, each move adds 2: 1 → 3 → 5 → 7 = BL. *Concept:* positional series · *Diff:* M · *Why NID:* tracking · *Mistake:* counting only the corners.

VR-116. Analogy: a line is to a square as a curve is to a:
(a) circle (b) cube (c) point (d) triangle
Ans: (a). Closing straight lines makes a square; closing a curve makes a circle. *Concept:* element-to-shape analogy · *Diff:* E · *Why NID:* design vocabulary · *Mistake:* none.

VR-117. Series of stick-figure frames: standing → crouching → mid-air with arms up → ? Most logical next frame:
(a) lying down asleep (b) landing with bent knees (c) standing with arms crossed (d) sitting on a chair
Ans: (b). The sequence shows a jump, and landing completes the action. *Concept:* sequential logic · *Diff:* E · *Why NID:* storyboard thinking · *Mistake:* none.

VR-118. Odd one out among logos: Olympic rings, Audi rings, MasterCard circles, Nike swoosh.
(a) Olympic (b) Audi (c) MasterCard (d) Nike
Ans: (d). The other three are built from overlapping or interlocking circles. *Concept:* visual classification · *Diff:* E · *Why NID:* logo awareness · *Mistake:* none.

VR-119. Series of growing squares: 1×1, 2×2, 3×3, 4×4 grids of unit tiles. How many unit tiles are added to go from the 4×4 to the 5×5?
(a) 5 (b) 9 (c) 10 (d) 25
Ans: (b). 25 − 16 = 9 (an L-shaped band). *Concept:* visual series (odd numbers) · *Diff:* M · *Why NID:* pattern · *Mistake:* answering 25.

VR-120. Analogy: footprint : foot :: shadow : ?
(a) light (b) object (c) sun (d) darkness
Ans: (b). Each is an impression or projection of the thing. *Concept:* relationship · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering "sun", which is the cause rather than the source shape.

G. Orthographic Views, Shadows & Perspective (121–135)

VR-121. The top view (plan) of a cone standing on its base is:
(a) a triangle (b) a circle with a dot at the centre (c) a square (d) an ellipse
Ans: (b). *Concept:* orthographic views · *Diff:* E · *Why NID:* 3D→2D · *Mistake:* answering triangle, which is the front view.

VR-122. The front view of a cylinder lying horizontally (its axis left–right) is:
(a) a circle (b) a rectangle (c) a triangle (d) an ellipse
Ans: (b). *Concept:* views · *Diff:* E · *Why NID:* basics · *Mistake:* answering circle, which is the end (side) view.

VR-123. The top view of a square pyramid is:
(a) a triangle (b) a square with both diagonals drawn (c) a circle (d) a pentagon
Ans: (b). The four sloping edges meet at the apex above the centre. *Concept:* views · *Diff:* M · *Why NID:* standard · *Mistake:* answering square only.

VR-124. A cube has a cylindrical hole drilled straight through from its front face to its back face. In technical drawing, the top view shows:
(a) a square with a circle (b) a square with two parallel dashed lines (c) a plain square (d) a circle only
Ans: (b). The hole is hidden inside, so its edges appear as dashed lines. *Concept:* hidden lines · *Diff:* H · *Why NID:* technical visualisation · *Mistake:* answering (a), which is the front view.

VR-125. The front view of a hemisphere resting on its flat face is:
(a) a circle (b) a semicircle (c) a square (d) a triangle
Ans: (b). *Concept:* views · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-126. At 8 am in India, a person faces their shadow. Which direction are they facing?
(a) East (b) West (c) North (d) South
Ans: (b). The morning sun is in the east, so the shadow points west. *Concept:* shadows and directions · *Diff:* E · *Why NID:* the classic item · *Mistake:* answering East.

VR-127. In a drawing, the light source is at the top-left. Where should the cast shadow of a ball on the ground fall?
(a) top-left (b) bottom-right of the ball (c) directly above (d) left
Ans: (b). Shadows fall away from the light. *Concept:* light logic · *Diff:* E · *Why NID:* rendering · *Mistake:* none.

VR-128. At what time of day are outdoor shadows shortest?
(a) Sunrise (b) Around solar noon (c) Sunset (d) Midnight
Ans: (b). The sun is highest then. *Concept:* shadow length · *Diff:* E · *Why NID:* scene realism · *Mistake:* none.

VR-129. In one-point perspective, lines that recede from the viewer meet at:
(a) two points on the horizon (b) one vanishing point on the horizon line (c) the bottom of the page (d) they never meet
Ans: (b). *Concept:* perspective · *Diff:* E · *Why NID:* scene drawing · *Mistake:* none.

VR-130. In a street scene at standing eye level, a distant standing adult's head should be drawn:
(a) above the horizon (b) on the horizon line (c) below the horizon (d) at the vanishing point only
Ans: (b). *Concept:* figures in perspective · *Diff:* M · *Why NID:* crowd scenes · *Mistake:* drawing far-off people as tiny figures below the horizon.

VR-131. You look down from the top of a tall building. Which perspective fits best?
(a) One-point at eye level (b) Three-point with the third vanishing point below (c) Isometric (d) Two-point with a low horizon
Ans: (b). *Concept:* bird's-eye view · *Diff:* M · *Why NID:* viewpoint tasks · *Mistake:* none.

VR-132. Isometric drawings differ from perspective drawings because:
(a) parallel edges remain parallel and nothing converges (b) they use 3 vanishing points (c) far objects look smaller (d) they are always 2D
Ans: (a). *Concept:* projection types · *Diff:* M · *Why NID:* product sketching · *Mistake:* none.

VR-133. In isometric projection, the three axes are drawn at:
(a) 90° to each other (b) 120° to each other on paper (with the receding axes at 30° from horizontal) (c) 45° (d) 60°
Ans: (b). *Concept:* isometric · *Diff:* M · *Why NID:* technical sketching · *Mistake:* answering 45°, which belongs to oblique / cabinet projection.

VR-134. [VISUAL QUESTION — IMAGE REQUIRED] *Description:* a silhouette shows a rounded body, a spout on the right curving upward, a looped handle on the left and a small knob on top. Which object is it?
(a) Watering can (b) Teapot / kettle (c) Pressure cooker (d) Vase
Ans: (b). A spout, a handle and a lid knob together identify a teapot or kettle. A watering can usually has a long straight spout and a top handle. *Concept:* silhouette recognition · *Diff:* E · *Why NID:* observation · *Mistake:* answering watering can.

VR-135. The side view of a standard mug (handle on the right, seen from the side) is roughly:
(a) a circle (b) a rectangle with a C-shape attached on the right (c) a triangle (d) an ellipse only
Ans: (b). *Concept:* orthographic views of everyday objects · *Diff:* E · *Why NID:* product drawing · *Mistake:* none.

H. Spatial Relations, Gears, Maps & Mixed (136–150)

VR-136. Gear A (turning clockwise) meshes with Gear B, which meshes with Gear C, in a line. Which way does Gear C turn?
(a) Clockwise (b) Anticlockwise (c) It doesn't move (d) It alternates
Ans: (a). Meshing gears alternate direction: A clockwise, B anticlockwise, C clockwise. *Concept:* mechanical reasoning · *Diff:* E · *Why NID:* product mechanisms · *Mistake:* none.

VR-137. Three gears are arranged in a closed triangle, each meshing with the other two. What happens when one is turned?
(a) All turn clockwise (b) They jam / cannot turn (c) They alternate (d) Only one turns
Ans: (b). An odd closed loop forces a direction conflict. *Concept:* mechanical logic · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

VR-138. Gear A has 20 teeth and drives Gear B with 60 teeth. If A turns 6 times, how many times does B turn?
(a) 18 (b) 2 (c) 3 (d) 6
Ans: (b). Teeth passed = 20 × 6 = 120, and 120 ÷ 60 = 2. *Concept:* gear ratio · *Diff:* M · *Why NID:* mechanism thinking · *Mistake:* answering 18.

VR-139. Two pulleys are connected by an open (uncrossed) belt. If the first turns clockwise, the second turns:
(a) clockwise (b) anticlockwise (c) it stays still (d) randomly
Ans: (a). An open belt keeps the same direction; a crossed belt reverses it. *Concept:* pulleys · *Diff:* E · *Why NID:* mechanisms · *Mistake:* none.

VR-140. You face North, turn right, turn right again, then turn left. Which direction do you face?
(a) North (b) East (c) South (d) West
Ans: (b). N → E → S → E. *Concept:* direction sense · *Diff:* E · *Why NID:* spatial orientation · *Mistake:* none.

VR-141. On a map, North is up. A temple is 3 km east and 4 km north of a school. What is the straight-line distance?
(a) 5 km (b) 7 km (c) 1 km (d) 12 km
Ans: (a). A 3-4-5 right triangle. *Concept:* map distance · *Diff:* E · *Why NID:* spatial reasoning · *Mistake:* answering 7.

VR-142. A person walks 5 m forward, turns 90° left and walks 5 m, turns 90° left and walks 5 m, then turns 90° left and walks 5 m. Where are they?
(a) 5 m from the start (b) Back at the start (c) 10 m away (d) 20 m away
Ans: (b). They have walked a closed square. *Concept:* path tracing · *Diff:* E · *Why NID:* basics · *Mistake:* none.

VR-143. In a row of 5 objects, a cup is 2nd from the left and a vase is 2nd from the right. How many objects lie between them?
(a) 0 (b) 1 (c) 2 (d) 3
Ans: (b). The cup is at position 2 and the vase at position 4, so position 3 lies between. *Concept:* positional reasoning · *Diff:* E · *Why NID:* logic · *Mistake:* none.

VR-144. A ladder leans against a wall with its top at 4 m and its foot 3 m from the wall. How long is the ladder?
(a) 5 m (b) 7 m (c) 6 m (d) 4.5 m
Ans: (a). *Concept:* spatial geometry · *Diff:* E · *Why NID:* proportion in drawings · *Mistake:* none.

VR-145. A lever: a 10 kg load sits 1 m from the fulcrum. What force is needed at 2 m on the other side to balance it?
(a) 5 kg-force (b) 10 kg-force (c) 20 kg-force (d) 2 kg-force
Ans: (a). Moments balance: 10 × 1 = F × 2, so F = 5. *Concept:* levers · *Diff:* M · *Why NID:* product mechanisms (bottle openers, seesaws) · *Mistake:* none.

VR-146. A bicycle's front gear (chainring) has 48 teeth and the rear sprocket has 16. For each pedal turn, the rear wheel turns:
(a) 1/3 time (b) 3 times (c) 16 times (d) 48 times

Ans: (b). 48 ÷ 16 = 3. *Concept:* gear ratio · *Diff:* M · *Why NID:* product understanding · *Mistake:* inverting the ratio.

VR-147. Which knot or join holds best when two sticks must be tied at right angles for a bamboo frame?
(a) A reef knot (b) Square lashing (c) A bow (d) A slip knot
Ans: (b). Square lashing is designed for right-angle joins in pole structures. *Concept:* structural making (Studio Test) · *Diff:* M · *Why NID:* hands-on making · *Mistake:* none.

VR-148. Which structure is strongest for a paper bridge using the same amount of paper?
(a) A flat sheet (b) A sheet folded into a corrugated (zig-zag) pattern (c) A crumpled ball (d) A loosely rolled tube with gaps
Ans: (b). Corrugation increases stiffness by moving material away from the bending axis. *Concept:* structural form · *Diff:* M · *Why NID:* Studio Test challenges · *Mistake:* answering (a).

VR-149. Triangles are used in bridge trusses and roof frames because:
(a) they look nice (b) a triangle cannot change shape without changing a side length, so it is rigid (c) they use more material (d) squares are illegal
Ans: (b). *Concept:* structural rigidity · *Diff:* E · *Why NID:* structures and models · *Mistake:* none.

VR-150. [VISUAL QUESTION — IMAGE REQUIRED] *Description:* four identical circles have their centres at the corners of a square. Neighbouring circles (along the square's sides) overlap. The two diagonal pairs of circles just touch at the centre of the square, and all four circles pass through that centre point. How many distinct enclosed regions are formed (not counting the outside)?
(a) 8 (b) 9 (c) 12 (d) 13
Ans: (a). There are 4 lens-shaped overlaps, one between each pair of neighbouring circles. No point lies inside three circles at once, because the diagonal circles only touch. Each circle also keeps one non-overlapping region, giving 4 more. All four lenses meet at the centre point, so there is no separate central region. Total 4 + 4 = 8. *Concept:* region counting · *Diff:* H · *Why NID:* careful visual analysis · *Mistake:* inventing a central region (there would be one only if the circles overlapped more). (Draw it to confirm.)

Notes on items with special answers (please read)

- All items have exactly one correct answer. Items VR-22, 30, 47, 48, 52, 58, 79, 95, 108 and 150 were tightened during the answer-key audit to remove ambiguity.
- For fold, cut and rotation items, physically checking with a sheet of paper is the fastest way to build accuracy.

NEXT: Volume 4C — Logical Reasoning (100 questions) (`04C-VOL4-QB-LOGICAL-REASONING.md`)